

# Designing Emergent Behavior in Shape-Changing Robot Swarms: Organizational Differentiation, Embodiment, and Human Intervention

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A thesis submitted in partial fulfillment of the requirements for  
the degree of

Master of Science in Computational Design

at

Carnegie Mellon University

Pittsburgh, Pennsylvania

December 15, 2025

# Abstract

Emergent behavior in complex systems arises from local interactions and distributed feedback without centralized control, producing rich global organization. While extensively studied in natural systems and simulations, emergence in designed robotic systems remains difficult to intentionally shape, compare, and evaluate. In particular, how organizational structure, physical embodiment, and human intervention jointly influence emergent swarm behavior is still poorly understood.

This thesis investigates how organizational differentiation and feedback policies, embodied robotic constraints, and human-in-the-loop cybernetic control can be designed to modulate the complexity, adaptability, and legibility of emergent behaviors in robotic swarms. Focusing on modular and reconfigurable robotic systems, the research frames emergence not as a byproduct of control algorithms, but as a tunable behavioral outcome shaped by system organization and interaction structure.

The work combines digital swarm simulations, physical prototyping of reconfigurable modules, and human-interaction experiments to systematically explore how variations in communication topology, role differentiation, and intervention modalities affect emergent patterns. A set of measurable behavioral metrics, including pattern diversity, adaptability, robustness, and human interpretability, is proposed to evaluate emergent complexity across configurations.

By establishing a comparative evaluation framework for emergence in robotic swarms, this research contributes a design-oriented understanding of how complex behaviors can be intentionally structured, intervened upon, and assessed in humanrobot collectives.

# Acknowledgments

I would like to thank...

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# 1 Introduction

## 1.1 Introduction

Emergent collective behaviorsuch as flocking birds, ant colonies, or cellular tissuesoften arises without centralized control. Instead, complex global patterns emerge from many agents following simple local rules. This phenomenon, commonly referred to as *emergence*, has been widely studied in fields such as biology, physics, and complex systems science. In recent years, emergence has also become an important aspiration in robotics and interactive systems, where designers seek to build collectives of simple agents that can adapt, reorganize, and respond to changing environments.

As robotic systems increasingly operate in open-ended, uncertain, and human-centered contexts, centralized control and exhaustive pre-programming become impractical. Swarm robotics, modular robotics, and soft robotic systems promise scalability and adaptability, yet they also raise a fundamental challenge: how to design systems whose behavior is not explicitly scripted, but emerges through interaction. This thesis addresses this challenge by treating emergence not merely as a phenomenon to be observed, but as a design problem to be systematically explored.

### 1.1.1 Emergence as a Design Problem

Emergence is often celebrated for its aesthetic richness and adaptive potential, but it also poses a difficulty for engineering and design. By definition, emergent behavior cannot be fully predicted or controlled from the behavior of individual agents alone. While this indeterminacy is a strength in natural systems, it complicates the design of robotic collectives intended to function reliably in real-world environments.

From a design perspective, the central question is not whether emergence can occur, but under what conditions it arises, how its properties vary, and how it can be shaped without eliminating its open-ended character. As robotic systems expand toward architectural scale, adaptive materials, and long-term human interaction, understanding how emergent behavior



can be intentionally influenced becomes increasingly important. Designing for emergence therefore represents a shift from prescribing outcomes to designing the conditions under which behaviors unfold.

### 1.1.2 Limits of Current Swarm Robotics Approaches

Despite extensive research in swarm robotics and multi-agent systems, current approaches tend to emphasize algorithmic control and optimization. Many systems rely on fixed local policies, centrally tuned parameters, or learning-based controllers that aim to produce specific target behaviors. While effective in controlled settings, such approaches often struggle to generalize, adapt, or scale when system morphology, environment, or human involvement changes.

At the same time, advances in soft robotics and modular robotic systems highlight the role of physical embodiment in shaping behavior. Material properties, geometry, actuation mechanisms, and connectivity can constrain and enable interactions in ways that reduce control complexity. However, embodiment is frequently treated as an engineering constraint rather than as an explicit design variable for shaping collective behavior.

Human-robot interaction research further complicates this picture. While humans are commonly involved as supervisors, operators, or collaborators, they are rarely considered as participants within the feedback loops that govern emergent swarm behavior. As a result, the interaction between organizational structure, embodied constraints, and human intervention remains underexplored.

Taken together, these limitations point to a gap: although emergent behavior can be produced in robotic swarms, there is limited understanding of how it can be systematically shaped through design choices across organization, embodiment, and human interaction.

### 1.1.3 Research Questions and Hypotheses

As shown in Figure 1.1, emergent swarm behavior is shaped through a closed-loop interaction between organizational policies, embodied constraints, and human feedback.

This thesis investigates how emergent behavior in shape-changing robot swarms can be intentionally influenced through design. Rather than optimizing a single controller or task performance metric, the research focuses on how design decisions alter the richness, adaptability, and legibility of collective behavior.

The central research question is:

*How can differentiated organization and local policies increase the complexity of emergent behavior in shape-changing robot swarms?*

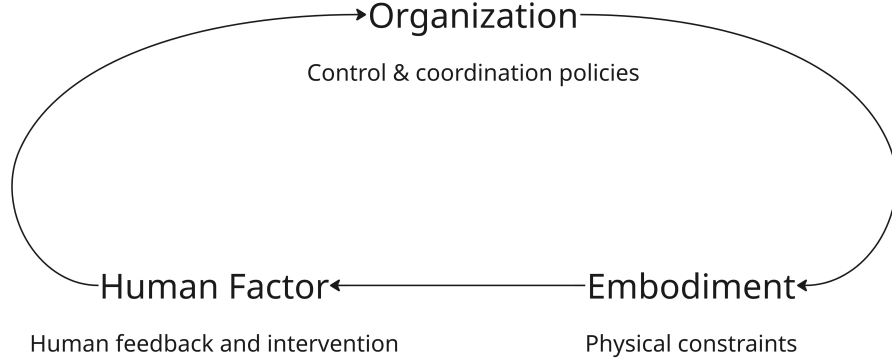


Figure 1.1: **Conceptual-dimensions** Three coupled dimensions shaping emergent behavior in robot swarms.

To address this question, the thesis advances two primary hypotheses:

**H1 Organizational Differentiation Hypothesis.** Introducing role differentiation within an otherwise homogeneous swarmsuch as variations in sensitivity thresholds, communication ranges, or actuation roleswill produce more diverse and adaptive emergent behaviors than a fully identical swarm.

**H2 Morphological Modularity Hypothesis.** Introducing morphing capabilities at the module levelsuch as inflatable structures or expanded connection conditionswill increase the range of physically realizable interactions, leading to a measurable increase in swarm behavioral complexity.

Together, these hypotheses frame emergence as an outcome shaped by organizational structure and embodied design, rather than solely by control algorithms.

### 1.1.4 Research Structure

This thesis is structured to systematically investigate how emergent behavior in shape-changing robot swarms can be shaped through organizational design, embodied constraints, and human intervention. The research progresses from conceptual grounding, to formal modeling, and finally to empirical evaluation across multiple experimental modalities.

**Chapter 2: Background Review** synthesizes prior work across three foundational domains that inform this study: emergent behavior in complex systems, swarm and modular robotics, and human–robot interaction. Rather than providing an exhaustive survey, this chapter focuses on identifying conceptual gaps and tensions between these fields—particularly the lack of design-oriented frameworks for shaping emergent behavior. The review establishes the theoretical motivation for treating emergence as a controllable design outcome rather than an incidental system property.

**Chapter 3: Models** is the design research methodology. [1] Reference Model and Impact Model formalizes the conceptual framework of the thesis. A reference model is introduced to describe how designable input factors—spanning organization, embodiment, and human interaction—mediate system dynamics and lead to observable behavioral outcomes. Building on this framework, an impact model derives concrete research hypotheses linking organizational differentiation and morphological modularity to measurable changes in emergent swarm complexity. These models serve as the analytical backbone connecting theory, design decisions, and experimental evaluation.

**Chapter 4: Methods** describes the methodological approach used to test the proposed hypotheses. The chapter outlines three complementary methods:

- (1) digital swarm simulations for controlled and repeatable comparison across organizational and morphological configurations,
- (2) physical prototyping of modular, shape-changing robotic units to validate embodied effects under real-world constraints,
- (3) exploratory human-in-the-loop evaluations to examine how human intervention influences legibility, adaptability, and behavioral transitions.

Together, these methods enable cross-validation between simulation, embodiment, and interaction.

**Chapter 5: Experiments and Results** presents the experimental studies and their outcomes. Results are organized according to the hypotheses introduced in Chapter 3, comparing emergent behaviors across different organizational structures and embodied configurations. Quantitative measures of behavioral complexity are complemented by qualitative observations of pattern formation, adaptation, and robustness. This chapter emphasizes comparative analysis across conditions rather than isolated demonstrations.

**Chapter 6: Discussion** interprets the experimental findings through the lens of the reference and impact models. The discussion reflects on how organizational differentiation, embodied constraints, and human intervention jointly shape emergent behavior, highlighting design trade-offs, limitations, and unexpected dynamics revealed by the experiments. Broader implications for swarm robotics, embodied intelligence, and human-machine co-adaptive sys-

tems are also discussed.

**Chapter 7: Conclusion and Future Work** summarizes the contributions of the thesis and outlines directions for future research. These include scaling toward larger swarms, integrating adaptive or programmable materials, refining evaluation metrics for emergent complexity, and exploring long-term humanswarm co-adaptation.

Overall, this structure supports a design-oriented investigation of emergence, moving from conceptual framing to empirical validation while maintaining a clear connection between design decisions, system dynamics, and behavioral outcomes.

## 2 Background and Literature Review

### 2.1 Dialectical Framing: From Opposition to Co-evolution

To situate this research, I begin not with a singular definition, but with a series of tension-spairs of concepts that appear contradictory yet are in fact mutually constitutive. Rather than resolving these oppositions, this thesis treats them as productive dialectics: dynamic relationships that generate new forms of thinking, designing, and computing.

#### Natural vs. Artificial

Everything originates from nature, yet human intelligence has introduced a peculiar rupture: the capacity to *produce artifacts*. This marks the emergence of the artificial not as something outside nature, but as a recursive extension of it.

Antoni Gaudí's claim *straight lines belong to men, curves to God* reveals a deeper distinction: artificial systems historically favor abstraction, regularity, and control, while natural systems embody continuous variation, adaptation, and emergence.

However, this boundary is increasingly unstable. Contemporary bio-inspired design, synthetic biology, and soft robotics suggest that artificial systems are no longer merely imposed upon nature but can *grow, adapt, and behave* like natural ones. The artificial, in this sense, becomes a *second-order nature*—a designed ecology of behaviors and materials.

#### Analog vs. Digital

Before digital computation, machines operated through continuous physical processes—voltages, rotations, and fluid flows. Analog systems compute through *physical equivalence*: the system itself embodies the phenomenon it models.

The tide prediction machine is emblematic: instead of calculating tides symbolically, it *physically simulates* them through gears and pulleys. Computation here is not abstract—it is material, embodied, and inseparable from physics.

Digital computation, by contrast, abstracts the world into discrete symbols (0/1), enabling scalability, precision, and programmability. Yet this abstraction comes at a cost: it separates computation from materiality.

Today, we observe a return toward hybrid paradigms, where computation is redistributed between digital control and physical form. This shift suggests that the analog/digital divide is not binary but a spectrum of *where computation happens*: in code, in matter, or across both.

## Predictable vs. Unpredictable

Digital systems privilege predictability. Discrete states, deterministic logic, and formal rules make behavior controllable and optimizable. This aligns with top-down design paradigms: define rules, execute processes, and predict outcomes.

In contrast, natural and many analog systems exhibit unpredictability not as noise, but as *emergence*. Systems such as reaction-diffusion, slime mold growth, or flocking behaviors generate complex global patterns from simple local interactions. These are bottom-up processes, where outcomes cannot be fully prescribed in advance.

Importantly, unpredictability is not the opposite of computation; it represents a different mode of it. It reflects systems where computation unfolds through interaction rather than instruction.

### 2.1.1 Synthesis: From Dialectics to Design Space

Across these three dialectics, a pattern emerges:

- Natural ↔ Artificial: from separation to **co-evolution**
- Analog ↔ Digital: from substitution to **distribution of computation**
- Predictable ↔ Unpredictable: from control to **negotiation with emergence**

These shifts collectively point toward a new paradigm: **computation is no longer confined to symbolic processing, but extends into material, morphological, and behavioral domains.**

This aligns with the notion of **morphological computation**, where physical form () participates in information processing. Instead of treating the body as passive and control as centralized, intelligence is distributed across structure, material, and environment. Such an approach challenges the dominance of top-down control and foregrounds the co-evolution of form and behavior.

### 2.1.2 Toward Reconfigurable Morphological Systems

Within this expanded computational paradigm, **reconfigurable soft robotic systems** emerge as a particularly compelling site of investigation.

- From the natural–artificial dialectic, they inherit the capacity to behave like living systems adaptive, compliant, and environment-responsive.
- From the analog–digital dialectic, they embody hybrid computation: material deformation (analog) coupled with sensing, learning, and control (digital).
- From the predictable–unpredictable dialectic, they operate at the edge of control where behavior is not fully scripted but shaped through interaction and feedback.

In particular, **inflatable modular robotics** introduces an additional layer: *reconfigurability through structure*. Inflation enables reversible, lightweight, and continuous morphological change, while modularity allows discrete recombination. Together, they form a system that navigates between continuity and discreteness, stability and transformation.

Such systems are not merely robots—they are **computational morphologies**: machines whose intelligence arises from the interplay between **shape (form)**, **matter (materiality)**, and **algorithm (control)**.

## 2.2 Emergent Behavior in Complex Systems

**From local interaction to global organization.** Emergent behavior is commonly described as coherent global organization arising from local interactions among many components, without a centralized controller. Classic accounts emphasize distributed feedback, nonlinear interactions, and sensitivity to initial conditions as mechanisms through which macroscopic patterns form and persist [2, 3]. In this view, self-organization is not an added module of control, but an outcome of repeated local rules and coupling between agents and environment.

**Defining “complex systems” and “emergence.”** Despite broad usage, “complex system” and “emergence” remain contested terms. Ladyman et al. argue that many commonly cited features are too vague to be analytically useful, and highlight the need to distinguish between qualitative descriptions and operational measures [4]. Prokopenko et al. further note that miscommunication often stems from unclear relations between complexity, self-organization, emergence, and adaptation, motivating complexity measures that can serve as

a shared vocabulary [5]. Complementing these efforts, de Haan proposes a framework for comparing emergence across domains by re-centering the role of the observer and abstracting away from specific mechanisms [6].

**Gap: controllability in designed systems.** Much of the complex systems literature either studies naturally occurring phenomena or evaluates emergence in abstract computational models. While these works clarify how emergence arises, they less frequently address how emergence can be *tuned* in engineered artifacts: i.e., how designable parameters shape the reachable space of emergent behaviors, and how to compare emergent outcomes across different embodiments, organizations, and interventions using shared evaluation criteria [4, 5]. This gap becomes critical when emergence is treated as a design target rather than merely an observed phenomenon.

## 2.3 Emergence in Robotic Systems

Robotic swarms and modular robotic systems offer an engineering context where local interaction rules, sensing/actuation constraints, and communication topology can be explicitly designed. In these systems, emergence is not only a scientific object of explanation but also a practical concern: emergent behaviors must be robust, interpretable, and controllable enough to support task performance and human operation.

### 2.3.1 Soft Robotics and Morphological Computation

**Embodiment as a generator of behavior.** A central argument in embodied intelligence is that morphology and material properties can function as computational resources, offloading complexity from centralized control to body–environment dynamics. Pfeifer and Bongard articulate how bodies shape perception-action loops and enable adaptive behavior through morphology-dependent interactions [7]. Related design principles emphasize ecological balance and sensory-motor coordination, framing intelligence as an emergent property of coupled morphology, materials, environment, and control [8].

**Material-to-behavior mapping and reducing control complexity.** In soft robotics, compliance, damping, and distributed actuation can produce rich behaviors with comparatively simple control signals. For example, multifunctional fluidic units can embody sensing, actuation, and control within physical circuits, enabling self-synchronization and other passive emergent dynamics through mechanical coupling [9]. Such systems motivate a view of



“behavior” as partially embedded in material and geometric constraints, rather than purely encoded in software.

**Relevance to this thesis.** For swarm robotics, embodiment influences the *behavioral repertoire* available to local policies: what counts as a feasible interaction, what patterns are physically stable, and how quickly a swarm can adapt. This thesis treats embodiment not as a backdrop, but as a controllable layer that shapes emergent outcomes and interacts with organizational structure and human intervention [7, 8].

## Modular and Reconfigurable Robotic Systems

**Topology, connectivity, and reconfiguration.** Modular robotic systems (MRS) formalize reconfiguration as a core capability: agents can form and break connections, change topology, and adapt morphology to tasks and environments. Ahmadzadeh and Masehian provide a comprehensive synthesis of MRS methods spanning abstraction, planning, control, and synchronization, while also highlighting persistent constraints in connection strength, communication, and coordinated action [10]. Landmark systems such as M-Blocks demonstrate self-reconfiguration through momentum-driven lattice movements and magnetic bonding, illustrating the tight coupling between mechanical design and the space of achievable collective behaviors [11? ].

**Large-scale collective organization and self-assembly.** At swarm scale, programmable self-assembly demonstrates how large numbers of simple robots can produce complex global forms under local rules [12]. In parallel, work on programmable self-assembly in materials and colloidal systems frames “programmability” as the design of interaction rules and building blocks that reliably yield target structures [13]. These precedents provide both inspiration and caution: emergence can be achieved, but systematic comparison across different organizational designs and embodied constraints remains challenging.

**From fixed rules to learned/evolved policies.** Recent approaches use evolutionary and learning-based methods to discover distributed policies that yield emergent behaviors, shifting the focus from hand-designed rules to exploration of a behavior space. Rajbhandari and Sofge, for example, use evolutionary training to produce desired emergent group behaviors across different robot embodiments in simulation [14]. Related work explicitly analyzes emergent behaviors under indirect swarm control, further emphasizing the need for characterization methods beyond task success [15].

**Gap: comparable studies of organization and feedback.** Although prior work demonstrates that reconfiguration, connectivity, and local policies can yield complex swarm behaviors, there remains a lack of systematic, comparable studies that isolate how *organizational differentiation* (e.g., role specialization, hierarchy) and *feedback strategies* (e.g., local vs. broadcast communication) modulate emergent complexity, robustness, and adaptability across embodiments and tasks [10, 14]. This thesis addresses this gap by proposing a comparative evaluation framework that links designable organizational factors to measurable emergent outcomes.

### 2.3.2 Human Intervention and Cybernetic Control

**Humans as part of the behavioral loop.** In many real deployments, swarms are not fully autonomous: human operators observe swarm state, interpret emergent patterns, and intervene to steer collective behavior. Human-robot interaction research frames this as an ongoing challenge, especially when robot behavior is complex, adaptive, or difficult to explain [16]. For swarm settings, human understanding is not merely an interface problem; it shapes the feasibility of control and collaboration.

**Perception and legibility of emergent behaviors.** A key prerequisite for human influence is the ability to recognize and interpret emergent swarm behaviors under noise and time constraints. Walker et al. study human recognition of canonical emergent behaviors (rendezvous, flocking, dispersion), showing differences in recognition difficulty and sensitivity to viewing conditions [17]. These results motivate treating *legibility* and *interpretability* as measurable success criteria rather than informal user impressions.

**Human feedback as a driver of discovery.** Beyond recognition, human feedback can be integrated into the process of discovering and categorizing emergent behaviors. Mattson and Brown propose a human-in-the-loop approach that learns a similarity space over collective behaviors and combines novelty search with clustering to discover a richer taxonomy of emergent behaviors in simulation [18]. This line of work reframes “human intervention” from direct teleoperation to a cybernetic role: humans shape exploration objectives, preference models, and the evaluation of novelty.

**Gap: evaluation framework for how intervention changes complexity.** Although prior work studies perception, interaction, and human-guided discovery in swarms, there is still no widely adopted framework that quantifies how different intervention modalities change emergent behavior complexity over time—for example, whether intervention increases

behavioral diversity, improves robustness, or collapses complexity into more predictable patterns [17, 18]. This thesis targets this gap by linking intervention modalities to measurable changes in emergent behavior metrics, and by positioning humans as an explicit element in the feedback loop rather than an external operator.

# 3 Problem Framing and Models

## 3.0.1 Models as a Research Methodology

This thesis treats models not only as descriptive diagrams but as a research methodology for structuring design, experimentation, and evaluation. Following the Design Research Methodology (DRM) proposed by Blessing and Chakrabarti [1], the work uses explicit models to link designable inputs, mediated system dynamics, and measurable outcomes. The Reference Model synthesizes prior literature into a structured design space for emergent swarm behavior, while the Impact Model translates this structure into concrete interventions, hypotheses, and evaluation criteria. Together, these models guide experimental design, ensure comparability across conditions, and frame emergence as a designable and assessable outcome rather than a purely descriptive phenomenon.

### Reference Model: Structuring Emergence as a Design Problem

To synthesize the diverse strands of literature reviewed above, this thesis introduces a design-oriented reference model that frames emergent swarm behavior as a structured design problem rather than a purely descriptive phenomenon. Drawing from complex systems theory, swarm robotics, and humanrobot interaction, the model provides a shared conceptual structure for reasoning about how emergent behaviors arise, how they can be influenced through design, and how their outcomes can be evaluated.

As illustrated in Figure 3.1, the reference model is structured along two complementary dimensions.

**Input, Influence, and Output.** Along the vertical axis, the model distinguishes between **Input** factors, **Influence** mechanisms, and **Output** criteria. Input factors correspond to parameters that can be directly designed and specified prior to system deployment, such as communication topology, degree of organizational differentiation, module morphology, and connection conditions. Influence factors represent mediated system dynamics that emerge during operation, including feedback propagation, adaptability, pattern transition frequency,

and locomotion capability. Output criteria capture the observable qualities by which emergent behavior is evaluated, including behavioral complexity, robustness, and legibility.

This separation reflects a central insight from complex systems research: emergent behavior cannot be directly controlled, but the conditions under which it arises can be systematically shaped through design choices. By distinguishing designable inputs from mediated dynamics and evaluative outcomes, the model clarifies what aspects of emergence are controllable, what must be inferred through observation, and what constitutes meaningful research results.

**Human Factor, Organization, and Embodiment.** Along the horizontal axis, the model situates swarm behavior within a coupled socio-technical system. The **Human Factor** captures how humans observe, interpret, and intervene in swarm behavior, including sensitivity to human input and intervention modalities. The **Organization** domain describes how agents coordinate through local policies, communication structures, role assignments, and swarm scale. The **Embodiment** domain represents the physical and material constraints of the robotic modules, including morphing capability, connective ability between modules, and locomotion affordances.

Across these domains, the model highlights several mediating mechanisms emphasized in prior work. Feedback range and communication topology shape the speed and locality of information propagation; organizational differentiation determines how many distinct local rules coexist within the swarm; and embodied constraints delimit the action space available to local policies. Emergent behavior arises from the interaction of these mechanisms rather than from any single factor in isolation.

Overall, the reference model functions as a conceptual synthesis of the background literature, establishing a structured vocabulary for discussing emergence as a design outcome that is shaped but not centrally controlled by human, organizational, and embodied factors.

### Impact Model: Interventions, Hypotheses, and Evaluation Logic

While the reference model structures the design space, the impact model translates this structure into a causal research logic that links design interventions to measurable changes in emergent swarm behavior. In contrast to studies that present emergence as an observed side effect of control policies or learning algorithms, the impact model treats emergent complexity as an explicit outcome variable.

Figure 3.2 extends the reference model by explicitly mapping design interventions to mediated mechanisms and evaluative criteria.

**From emergence as description to emergence as outcome.** Much of the swarm robotics literature demonstrates emergent behavior through compelling examples, yet often lacks a framework for comparing how different design choices alter the *complexity* of those behaviors. The impact model addresses this gap by framing emergent complexity as a dependent variable that can be systematically compared across organizational and embodied configurations.

**Design interventions.** Two primary intervention dimensions are derived from the reference model. The first concerns **organizational differentiation**, including the introduction of heterogeneous agent roles within an otherwise homogeneous swarm. This may involve differences in sensitivity thresholds, actuation roles, or communication ranges, while maintaining a shared control architecture. The second concerns **morphological modularity**, in which robot modules gain shape-changing capabilities (e.g., inflatable or morphing structures) and more permissive connection conditions, expanding the set of physically realizable collective behaviors.

Rather than prescribing global behavior, these interventions act indirectly through mediated mechanisms such as feedback range, adaptability, and reconfiguration dynamics.

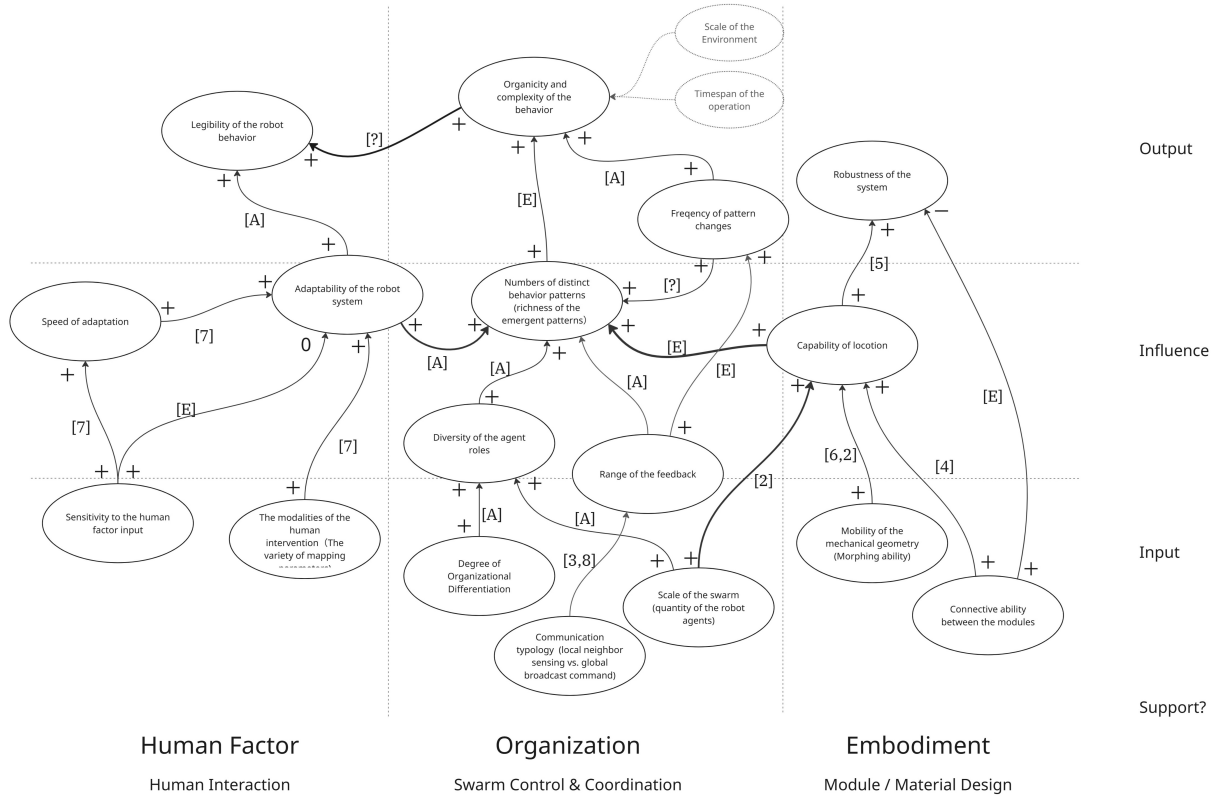
**Hypotheses.** Based on this causal structure, the thesis advances the following hypotheses:

**H1 Organizational Differentiation Hypothesis.** Introducing role differentiation within an otherwise homogeneous shape-changing swarm will produce a greater diversity of emergent behavior patterns and increased adaptability compared to a fully identical swarm operating under matched conditions.

**H2 Morphological Modular Geometry Hypothesis.** Enhancing the morphing ability and connection affordances of robot modules will lead to a measurable increase in emergent behavioral complexity, enabling a broader range of collective formations and improved robustness to environmental constraints.

**Evaluation logic.** To assess these hypotheses, the impact model emphasizes comparative evaluation across configurations rather than reliance on a single task-based metric. Emergent behavior is evaluated using a combination of behavioral richness, adaptability, robustness, and legibility, allowing different organizational and embodied designs to be analyzed on equal terms.

Together, the reference and impact models establish a coherent bridge between the background literature and the proposed methods, framing emergence as a designable, measurable, and comparable phenomenon in shape-changing robotic swarms.



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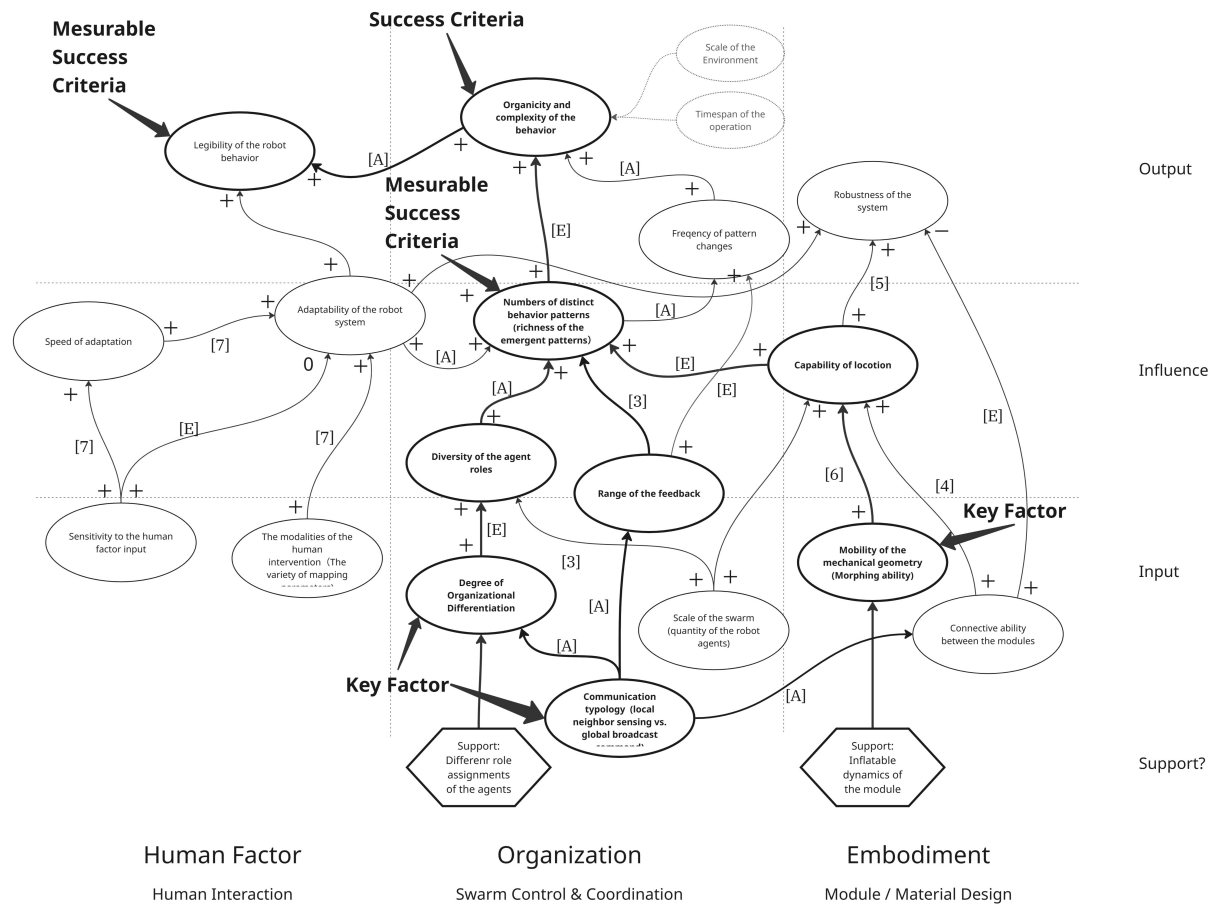
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[8] "Stephen Wolfram: A New Kind of Science | Online—Table of Contents." Accessed November 12, 2025.

Figure 3.1: **Reference model for emergent swarm behavior.** The model organizes design factors along two axes. Vertically, factors are structured from *Input* (directly designable parameters), through *Influence* (mediated system dynamics), to *Output* (success criteria of emergent behavior). Horizontally, the model spans *Human Factor*, *Organization*, and *Embodiment*, capturing how human intervention, swarm coordination, and physical module design jointly shape emergent outcomes.



**Figure 3.2: Impact model linking design interventions to measurable emergent outcomes.** The model specifies how interventions on organizational structure and embodied morphology influence mediated dynamics such as feedback propagation, role diversity, and reconfiguration. These influences are evaluated using measurable success criteria, enabling comparative analysis of emergent behavior across swarm configurations.



# 4 Methodology

## 4.1 Methods

This thesis investigates a design-oriented question: *how do organizational differentiation and local interaction policies, together with embodied modular morphing, increase the emergent behavioral complexity of a shape-changing robot swarm?* To answer this question, we construct a multi-stage experimental pipeline that couples

- (i) physics-based swarm simulation,
- (ii) mixed-reality human-in-the-loop interaction tests,
- (iii) physical prototyping, with a unified evaluation layer that operationalizes “emergent complexity” into measurable proxies.

Methodologically, the work treats emergence as a *mediated outcome*: designable inputs (module morphology, connection conditions, organizational structure, and human intervention channels) do not directly specify a target global behavior; instead, they shape the conditions under which global patterns appear, persist, and transition. This perspective is aligned with cybernetic framings of human-in-the-loop systems [16], and motivates an experimental design that emphasizes comparative regimes and perturbation-response analysis rather than single-demo success.

**Overview of experimental logic.** We implement a two-layer simulation stack: a *physical layer* in Isaac Sim that provides credible embodied constraints (contact, friction, connection forces), and an *interaction layer* in Unity that injects human intervention as an input channel while preserving first-person/immersive observation. We then translate key assumptions into physical prototypes to verify which emergent phenomena survive material uncertainty. Across all modalities, we evaluate outcomes using a shared set of metrics:

- (1) behavior repertoire and regime diversity,
- (2) transition dynamics and adaptability,
- (3) robustness under noise/perturbation,
- (4) human legibility and intervention effectiveness.



Figure 4.1: **Swarm robot behavior simulation** in the Unity platform with 2 different perspectives.

### 4.1.1 Simulation Development (Isaac Sim + Unity/MR)

#### Platform choice as “two-layer simulation”

We deliberately separate physics fidelity from interaction fidelity. Isaac Sim is used as the authoritative physics engine for multi-agent dynamics: rigid-body simulation, contact resolution, frictional interactions, and constraint-based connections are handled in a reproducible environment, allowing controlled trials and parameter sweeps. This choice supports modular robotics assumptions where the feasibility of motion and reconfiguration strongly depends on contact geometry and connection constraints [10?, 11].

Unity is used as the human-facing interaction and visualization layer. As shown in the 4.1 it provides first-person/immersive observation, configurable user input mappings, and logging of human actions and subjective judgments. Crucially, the Unity layer does *not* replace physics; instead, it acts as a mediated control surface that sends high-level interventions (parameter modulation, triggers, or spatial cues) into the Isaac simulation. This “two-layer” design directly supports the thesis framing of *humans as part of the behavioral loop* [16]: humans observe emergent states, intervene through constrained channels, and re-observe the resulting transitions.

## Module modeling via behavior primitives and connection rules

A recurring failure mode in swarm studies is that collective comparisons become unstable when individual agent capabilities are underspecified. To avoid this, we construct two explicit capability libraries for a single module: (i) a *shape-changing primitive library* describing what morphing actions exist and how they unfold over time, and (ii) a *connection rule library* describing when modules attach/detach and what constraints result. This abstraction strategy mirrors how modular robotics literature separates action capabilities from coordination algorithms [10] and how lattice-based systems formalize feasible reconfiguration moves [11].

**Shape-changing primitives (morphing).** Each module exposes a small set of parameterized morphing primitives such as inflate/deflate pulses, sustained expansion, and oscillatory actuation. Primitives are defined by time-series parameters (rise time, frequency, amplitude), and by their geometric and contact consequences (volume change, contact area change, passability through gaps). This operationalizes the morphological computation intuition that “body” and “material” contribute to behavior generation and reduce the burden of centralized control [7, 8].

**Connection rules (attachment constraints).** Connection is treated as a stateful constraint with measurable thresholds. We implement: (1) trigger conditions (distance threshold, orientation threshold, contact area threshold), (2) strength parameters (maximum tension/torque before failure), (3) detach logic (exceeding force thresholds or intentional release), and (4) topology constraints (multi-point attachment, maximum degree, and whether reconfiguration is allowed). This parameterization is motivated by the fact that modular systems often hinge on the tight coupling between kinematic feasibility and connection reliability [10? ].

**Deliverables of A2.** The output of this stage is not a “nice demo” but a calibrated capability specification: (i) repeatable primitive trajectories (morph curves over time), (ii) stable attach/detach boundaries (trigger regions in parameter space), and (iii) empirical distributions of failure modes (false attachments, premature detachments). These results directly seed the physical prototype design by providing initial parameter ranges for actuation bandwidth and connection tolerances.

### Coordination typology: scanning organization $\times$ local policy

The goal of coordination experiments is to make *organizational differentiation* explicit and comparable, so that the Organizational Differentiation Hypothesis (H1) can be tested against a homogeneous baseline. Rather than implementing a single canonical swarm algorithm, we define a *policy family* and evaluate how behavior regimes change across organization structures and communication topologies.

**Minimal comparative set (core controls).** We implement at least three conditions:

- **Homogeneous baseline:** all agents share identical sensing thresholds, morphing bandwidth, and connection preferences.
- **Differentiated roles:** at least two roles are introduced (e.g., explorer/anchor/relay), realized by different sensitivity thresholds, actuation bandwidths, or communication radii.
- **Topology variation:** local neighbor sensing versus broadcast-like information sharing (or sparse local graph versus global shared state).

This design isolates the effect of differentiation from the effect of communication structure, aligning with modular robotics perspectives that treat abstraction, coordination, and synchronization as distinct layers of the problem [10].

**Parameter sweep and regime discovery.** For each condition, we run parameter sweeps over a bounded design space: morph amplitude/frequency, connection thresholds/strength, neighborhood radius, and role proportion. Each run produces spatiotemporal trajectories: agent poses, velocity fields, connection graph time-series, and morph state sequences. The analysis goal is to discover *regimes* (recurrent collective patterns) and quantify how the regime landscape changes when organizational differentiation or morphing capability is introduced.

**Taxonomy building through behavior embeddings.** To avoid subjective cherry-picking of “interesting” behaviors, we represent each trial by an aggregated behavior descriptor (e.g., time-series statistics of graph connectivity, density, alignment, and morph state occupancy). We then cluster trials to form a taxonomy of emergent patterns and compute “richness” as the number and diversity of clusters. This taxonomy-based approach is consistent with the idea that one should explore and categorize the space of possible swarm behaviors rather than target a single engineered endpoint; it also parallels methods that combine novelty-driven exploration with clustering and similarity spaces [18].

#### A4. Mixed-reality human-in-the-loop experiments

Human-in-the-loop evaluation is included for two reasons. First, swarm behaviors are often used in contexts where a human operator must recognize the current collective mode and decide whether and how to intervene. Second, “organic” complexity has an irreducible perceptual component: behaviors that are mathematically complex may still be illegible or unusable for human supervision. Therefore, we separate two MR study aims and design tasks accordingly.

**Aim 1: legibility and recognition of emergent modes.** We adapt a recognition-style protocol: participants view short time windows of swarm behavior and categorize the observed mode (or detect whether a transition occurred), with confidence ratings and response times. This directly follows the premise that different emergent behaviors have different perceptual recognizability, and that recognition performance changes with noise and viewing time [17]. In our context, the key independent variables are organizational differentiation (homogeneous vs role-differentiated), morphing capability (enabled vs disabled), and visual noise/occlusion levels introduced in the MR display.

**Aim 2: intervention effectiveness under mediated control.** Participants intervene through constrained channels (e.g., adjusting a small set of policy parameters, triggering role switches, or injecting spatial cues). We log intervention frequency, type, and timing, then measure downstream changes in regime transitions, complexity metrics, and recovery dynamics. This design treats intervention as *indirect control* rather than direct command, aligning with frameworks that analyze emergent behaviors under indirect swarm control [15]. The study does not assume participants know which behaviors are possible or desirable beforehand; instead, it emphasizes the co-adaptive process of discovery and preference, echoing the notion that human feedback can guide exploration of novel emergent behaviors [18].

#### 4.1.2 Robot Design and Embodiment Variables

To investigate how embodiment shapes emergent behavior in shape-changing robot swarms, this thesis explicitly parameterizes robot design as a set of controllable experimental variables. Rather than treating the robotic platform as a fixed carrier of control policies, embodiment is modeled as an active contributor to system dynamics, consistent with perspectives from morphological computation and soft robotics. The selected variables define the design space of the modular inflatable units and are systematically explored across simulation and physical prototyping.

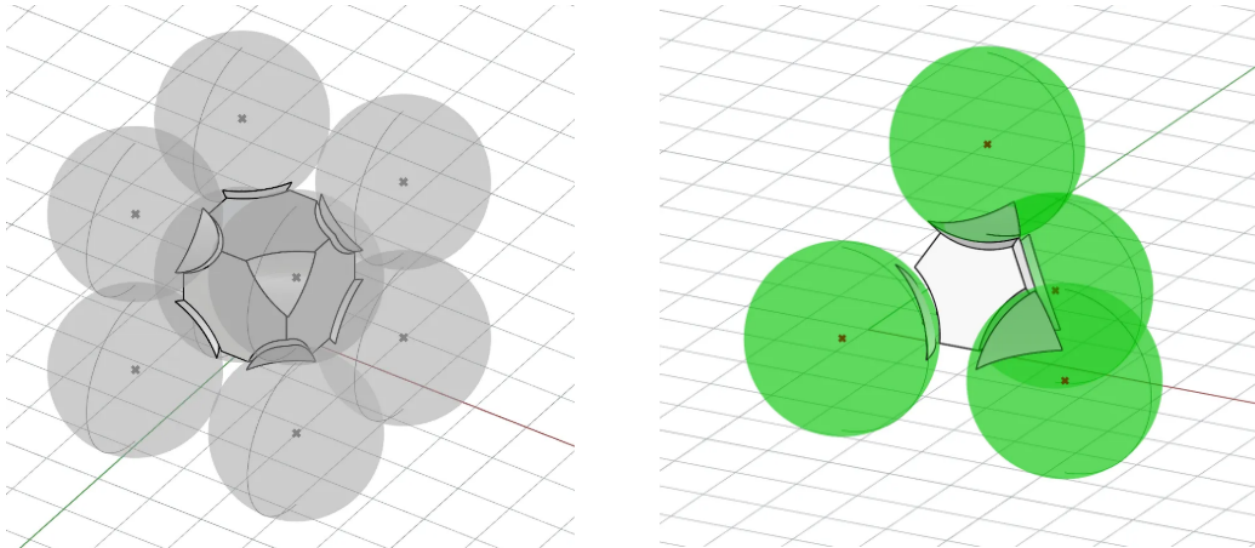


Figure 4.2: **Different geometries for the central connector.** Alternative connector geometries define distinct contact interfaces between modules, influencing connection stability, allowable relative motion, and force transmission. By varying connector shape while keeping actuation mechanisms constant, the study isolates geometric embodiment effects on swarm-level behavior.

**Core Geometry and Modular Topology.** The first variable concerns the geometry of the core connector that determines how many inflatable modules can be combined and how they are spatially arranged. Core geometries such as tetrahedral, cubic, or hybrid configurations define the maximum degree of connectivity, local symmetry, and feasible swarm topologies. This variable directly constrains how forces, deformations, and motion propagate through connected modules, influencing collective locomotion and reconfiguration capabilities.

The geometry of the central connector is treated as a primary embodiment variable. Different connector profiles introduce variation in mechanical compliance, alignment tolerance, and the effective degrees of freedom at inter-module joints (Figure 4.2). These geometric differences are expected to influence how local interactions propagate through the swarm, affecting pattern formation, locomotion modes, and the stability of emergent configurations.

**Shape-Changing Control Mechanisms.** Each module is capable of shape change through controlled inflation and deflation of multiple inflatable patches. Control parameters include inflation speed, temporal resolution (timesteps), amplitude, and phase differences between patches. In simulation, these parameters are instantiated as continuous control signals (e.g., sinusoidal phase offsets), forming a library of shape-changing primitives. Variations in these primitives affect rhythmic coordination, self-propelled motion, and coupling between neigh-

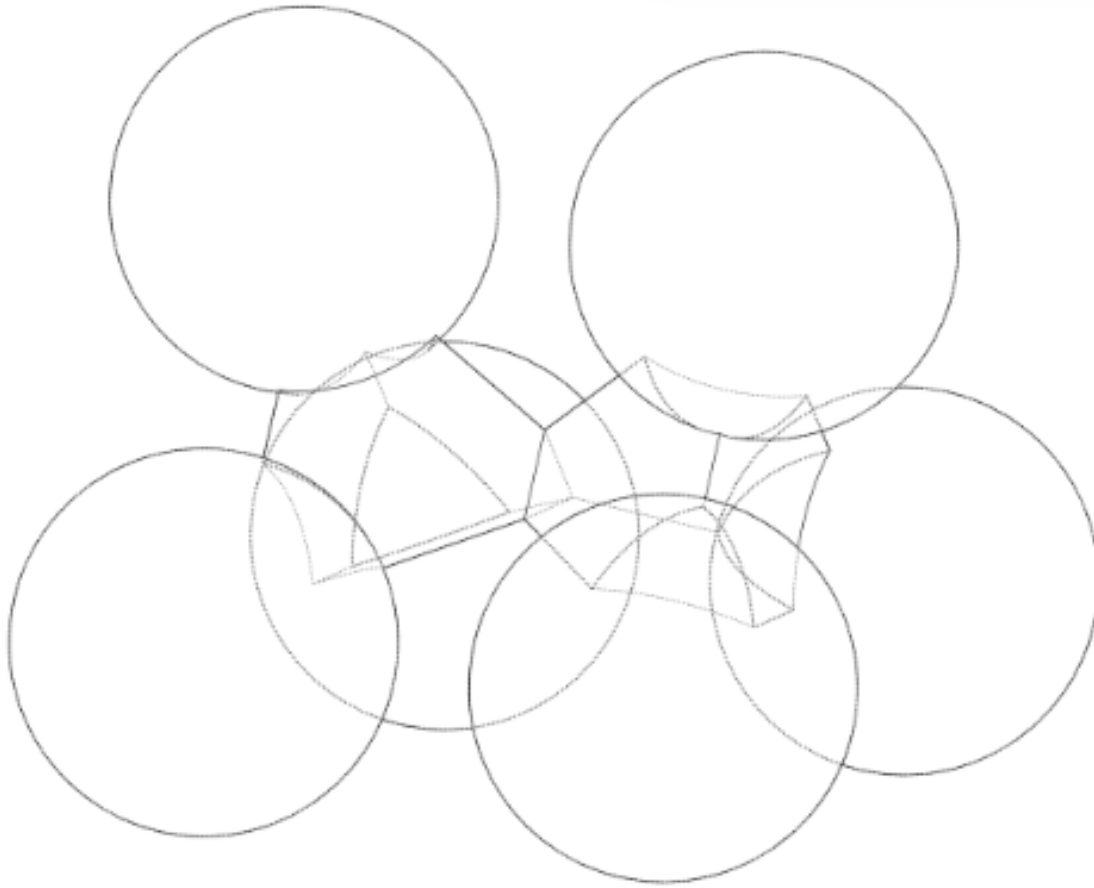


Figure 4.3: **Modular structure enabling reconfiguration.** The modular architecture allows units to form, break, and reorganize connections, enabling dynamic changes in swarm topology. This capability supports experimental manipulation of connectivity, role differentiation, and organizational structure.

boring modules.

**Connection Conditions and Reconfigurability.** Connection conditions define when and how modules attach or detach from one another, enabling reconfigurable swarm structures. These conditions include distance thresholds, orientation constraints, contact area requirements, and force or torque limits for maintaining connections. By varying these parameters, the system explores different degrees of structural flexibility, from rigid assemblies to highly dynamic, self-reorganizing configurations. This variable is central to examining how connectivity mediates transitions between behavioral modes.

As illustrated in Figure 4.3, this reconfigurable structure enables controlled variation of several key embodiment-related variables, including connection degree, local topology, and

the persistence of connections over time. These variables are directly linked to organizational factors in the Reference Model, such as degree of differentiation and communication pathways, while remaining grounded in physically realizable constraints.

**Mass Distribution and Weight Shifting.** Weight is treated as a controllable variable through internal redistribution of fluid or material within a module. By pumping liquid into different sections, the center of gravity of a unit can be shifted over time. This introduces coupling between shape change and gravitational effects, influencing stability, rolling behaviors, and collective balance, especially in uneven or constrained environments.

**Material Properties.** Material choice, such as silicone gels or polyethylene-based films, introduces differences in elasticity, damping, friction, and deformation limits. These properties affect how shape changes translate into motion and how energy is stored or dissipated during interaction. Material variables are initially abstracted in simulation and later validated through targeted material tests in physical prototypes.

**Relation to Experimental Design.** Together, these variables constitute the embodiment-related *input factors* in the Reference Model and are manipulated independently or in combination across experimental conditions. Their effects are observed through mediated system dynamics such as coordination patterns, locomotion capabilities, and reconfiguration frequency and evaluated using behavioral, informational, and human-centered metrics described in Section ???. This structured treatment enables embodiment to be studied not as an implementation detail, but as a first-class experimental factor in emergent swarm behavior.

### 4.1.3 Physical Prototyping (embodiment as evidence)

Simulation is necessary for regime discovery, but insufficient as evidence of embodied emergence. Physical prototypes are therefore used to test whether key phenomena persist under frictional uncertainty, actuation drift, leakage, and imperfect connections. Importantly, the prototype stage is scoped for *comparative validation* rather than maximal scale: the goal is to reproduce regime contrasts (homogeneous vs differentiated; morphing vs non-morphing) in a controllable small- $N$  setting.

#### Single-module validation: morphology–behavior mapping

We first validate that the simulated morphing primitives correspond to physically achievable actuation signatures. For each primitive, we measure shape change amplitude,



response time, repeatability across cycles, and actuation curves (e.g., pressure or energy proxy where available). This stage supports H2 by grounding the claim that morphing capability is a meaningful design input rather than a purely simulated affordance. If modules exhibit coupling-induced phenomena such as spontaneous synchronization, we document these as embodied dynamics rather than controller artifacts, consistent with observations that mechanically coupled fluidic units can yield emergent passive behaviors [9].

### **Connection validation: attach/detach boundaries and constraint effects**

Next, we validate connection conditions with two-module and small-chain tests. We measure connection success rate, failure modes, detach thresholds, and how connectivity constrains motion degrees of freedom. We explicitly compare fixed connections versus reconfigurable connect/disconnect conditions. This stage directly operationalizes the Embodiment-domain variables “connective ability” and “connection conditions” and tests whether the mediated variables (e.g., locomotion capability and transition frequency) shift as predicted by the Reference Model.

### **Small-swarm comparative experiments ( $N \approx 5\text{--}20$ )**

Finally, we run minimal comparative experiments with a small swarm where role differentiation is physically instantiated. Role differences may be implemented through sensing thresholds, actuation bandwidth, or connection preferences (e.g., a role that tends to anchor via stronger connections versus a role that explores via faster morphing cycles). The objective is not to replicate thousand-robot demonstrations, but to establish a controlled setting in which organizational differentiation is visible, repeatable, and measurable.

This stage also supports a broader design framing: rather than pushing controllability upward into centralized planning, the work explores how controllability can be distributed into building-block-level properties and interaction rules, resonating with directions that treat reconfigurable matter as a programmable substrate [19].

## **4.1.4 Evaluation: operationalizing “complexity” into measurable success criteria**

A central methodological challenge is that “organic complexity” is an abstract goal. We therefore evaluate success through three complementary metric groups aligned with the Output layer of the models: (C1) complexity and repertoire richness, (C2) adaptability and robustness, and (C3) human legibility and intervention effects. The emphasis is on using multiple partially independent measures so that conclusions do not hinge on a single proxy.

## Emergent complexity and behavioral repertoire

**Repertoire size and taxonomy richness.** Using the clustering-based taxonomy from A3, we compute the number of distinct behavior categories observed under controlled conditions, and how evenly trials populate those categories. This directly matches the measurable proxy “numbers of distinct behavior patterns.” We report cluster counts, long-tail occupancy (rare-but-recurrent modes), and between-cluster separability.

**Transition dynamics as complexity-in-motion.** We quantify how often and how predictably the swarm switches between modes: transition frequency, mean dwell time per mode, and transition reversibility. These measures align with the Reference/Impact Model variables “frequency of pattern changes” and “adaptability.” In addition to raw counts, we compute transition matrices to characterize whether differentiation increases the *variety* of reachable modes or merely increases instability.

**Information-theoretic descriptors (supporting evidence).** To strengthen claims beyond taxonomy counts, we compute information-theoretic measures on time-series features such as connectivity graph statistics, density fields, velocity alignment, and morph-state occupancy. This follows the premise that complexity, self-organization, and emergence can be discussed with an information-theoretic “dictionary” that improves comparability across systems [5]. These metrics are treated as supporting evidence rather than the sole definition of success.

## Adaptability and robustness under perturbation

Emergent systems are often valued not for a single pattern, but for their ability to remain functional under uncertainty. We therefore evaluate:

- **Perturbation recovery time:** time to return to a stable mode after external disturbances (agent removal, connection failures, pushes, terrain changes).
- **Performance under noise:** degradation curves of complexity proxies under sensing noise, latency, and message loss.
- **Graceful degradation:** how regime richness changes as a function of reduced agent count or impaired role capabilities.

These tests also reflect practical constraints in modular robotic systems where resources, communication, and connection strength limit real-world deployments [10].

## Human factors: legibility, controllability, and intervention impact

**Legibility and recognition accuracy.** From the MR recognition tasks, we compute recognition accuracy, response time, and confidence across conditions. We further examine how noise and viewing time affect recognition, following the logic that recognition differs by behavior type and perceptual conditions [17]. This metric group connects directly to the Output variable “legibility of robot behavior.”

**Intervention effectiveness as indirect control.** For intervention trials, we quantify the causal footprint of human input: intervention rate, input-to-response latency, and pre/post changes in regime transitions and complexity proxies. Because the intervention channel is mediated rather than direct, we interpret results in the context of indirect swarm control: human inputs shift conditions, and global behavior responds through emergent dynamics [15]. Where relevant, we analyze whether human input expands the behavioral repertoire (discovering new modes) or compresses it (stabilizing a subset of modes).

### 4.1.5 Experimental design summary: hypotheses and interventions

The full experimental design is organized to test two primary hypotheses through explicit interventions and controlled comparisons:

**H1: Organizational Differentiation Hypothesis.** Introducing role differentiation within an otherwise homogeneous swarm will yield a larger and more diverse emergent repertoire than the baseline. This is tested by comparing homogeneous versus differentiated-role configurations under matched embodiment primitives and matched environment settings, evaluating changes in taxonomy richness, transition structure, and robustness/legibility measures.

**H2: Morphological Modularity Hypothesis.** Introducing morphing ability and richer connection conditions at the module level will increase measurable behavioral complexity. This is tested by comparing morph-enabled versus morph-disabled modules (or restricted primitives), while holding organizational policy families constant, and then validating key contrasts in physical prototypes.

**Interventions.** Interventions occur at three levels: (1) *Design-level* interventions (morph primitives, connection rules), (2) *Organization-level* interventions (role differentiation, topology), and (3) *Human-level* interventions (MR input channel and timing). The evaluation

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framework is structured to detect not only improved complexity but also the trade-off space among complexity, robustness, and legibility.

# 5 Experiments and Results

Content below is the placeholder.

## 6 Discussion and Contributions

Content below is the placeholder.

## 7 Conclusion and Future Work

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# A Additional Materials

Add supplementary figures, tables, protocols, etc.